

Ecophysiological Aspects of the Woody Root Hemiparasite *Santalum acuminatum* (R. Br.) A. DC and its Common Hosts in South Western Australia

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The water and nitrogen relationships of the xylem-tapping root hemiparasite, quandong (*Santalum acuminatum*) and its principal hosts were examined at a series of sites in native coastal heathlands of south west Australia. Assessments based on densities of above-ground biomass, ground cover and frequencies of haustoria on host roots indicated that woody N₂ fixers (legumes and *Allocasuarina*) were principal hosts of *Santalum*. $\delta^{15}\text{N}$ values for shoot dry matter of component species suggested these N₂ fixers were strongly (70% or more) dependent on atmospheric N and that *Santalum* derived N principally from these species. Structural studies of haustoria showed the interface with host xylem to be almost entirely comprised of parenchymatous tissue. No luminal continuities were observed between xylem conducting elements of the partners. Formation and functional life of haustoria were closely coordinated with seasonal growth of hosts, with some haustoria surviving summer and overlapping functionally with new ones establishing in the next autumn. Transpiration and photosynthetic rates of the parasite were consistently less and water use efficiencies very similar to those of the principal host *Acacia rostellifera*. $\delta^{13}\text{C}$ values of foliage of *Santalum* and this host were similar, but large variations in $\delta^{13}\text{C}$ values for above-ground dry matter of parasite and hosts between study sites prevented evaluations of water stress or water-use-efficiency based on carbon isotope discrimination. Specific hydraulic conductivities of roots of *Santalum* were consistently lower than those of *Acacia*, a finding consistent with more conservative water use by the parasite than the host.

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Key words: *Santalum acuminatum*, root hemiparasite, ^{15}N discrimination, water relations, haustorial structure, root conductivity.

INTRODUCTION

The genus *Santalum* (Santalaceae) comprises nine species of xylem-tapping root hemiparasitic trees or shrubs native to open woodlands of the southern hemisphere. Certain species (e.g. *S. album* and *S. spicatum*) have been heavily exploited in the wild for sandalwood and are now being successfully cultivated on a limited scale (McKinnell, 1991). Of the six species native to Australia, the quandong (*S. acuminatum*), has been recently investigated (Tennakoon, Pate and Stewart, 1997) with respect to the ability of its haustoria to absorb and process various nitrogenous and other solutes from the xylem of a range of common natural hosts, and the data compared with similar information from other species of root hemiparasites (see Press, 1989; Press, Graves and Stewart, 1990; Stewart and Press, 1990; Press, Smith and Stewart, 1991; Seel *et al.*, 1992; Pate *et al.*, 1994; Press, 1995; Tennakoon and Pate, 1996a; Tennakoon, Pate and Fineran, 1997; Tennakoon and Pate, 1997).

While detailed understanding of haustorial metabolism and functioning is clearly essential for appreciating the qualities underlying the benefits which a root hemiparasite may gain from single hosts in pot or plot culture, such evaluations have little ecophysiological relevance unless interpreted against the complex interactive biology of

multiple parasitism typical of natural habitat conditions. Bearing this in mind, the present paper examines a range of biological and functional aspects of the quandong and its principal hosts in pristine coastal habitats of south western Australia. The study includes evaluations of the extents to which species are parasitized, examinations of the gross morphology and anatomy of the haustorium and use of $\delta^{15}\text{N}$ natural abundance signals of parasite and native hosts to assess possible extents of dependence on specific hosts for N. In addition to these studies, the water relations of parasite and hosts, as studied by $\delta^{13}\text{C}$ analyses of dry matter, were investigated and detailed examinations made of the seasonal CO₂ and transpiration exchanges and root hydraulic conductivity characteristics of *Santalum* and its principal natural host *Acacia rostellifera*. The studies run in parallel with recent investigations of the biology of another S. W. Australian woody hemiparasite, *Olax phyllanthi* (Olacaceae) in its native habitat (see Fineran, Ingerfeld and Patterson, 1987; Kuo, Pate and Davidson, 1989; Pate *et al.*, 1990a, b, c; Tennakoon *et al.*, 1997a).

MATERIALS AND METHODS

Analyses of the sites and determination of host species

The study region was located in the northern sand plains of south western Australia, 40 km north of Eneabba (29° 49' S, 115° 16' E), and comprised three separate habitats (Fig.

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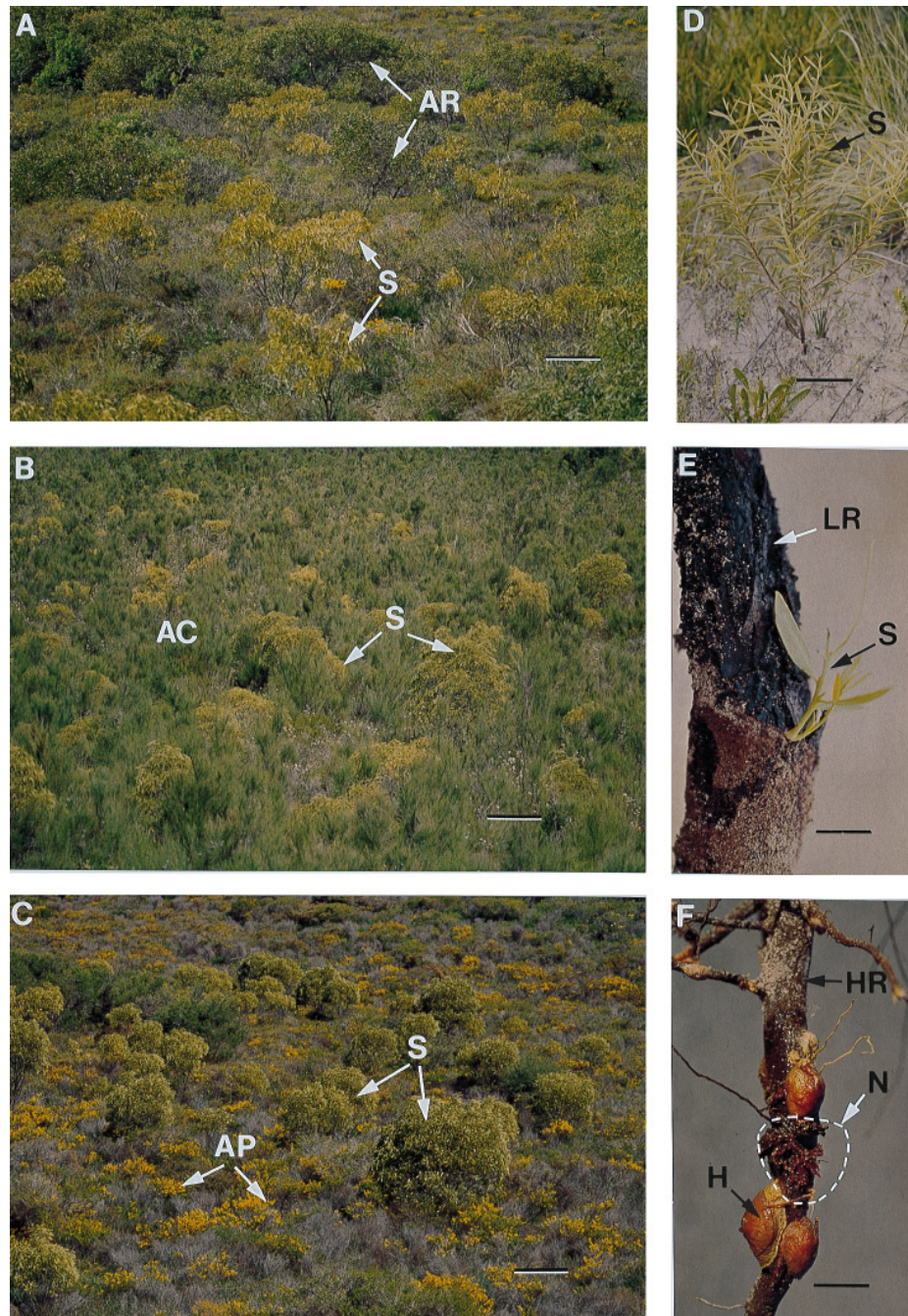


FIG. 1. General habitat views of *Santalum acuminatum* in northern sand plains at Eneabba, south-western Australia showing *Santalum* (S) in association with principal host species *Acacia rostellifera* (AR) at site A (A), *Allocasuarina campestris* (AC) at site B (B) and *Acacia pulchella* (AP) (yellow flowering) at site C (C). Bar = 1 m. D, *Santalum* plant (S) in natural habitat regenerating from a base of a burnt shoot. Bar = 10 cm. E, Resprouting shoot sucker of *Santalum acuminatum* (S) emanating from a lateral root (LR) following a mild fire. Bar = 2 cm. F, Lateral root of the host *Allocasuarina campestris* (HR) bearing fully mature *Santalum* haustoria (H). Note the nitrogen-fixing 'nodular' outgrowth (N) on the root of *Allocasuarina*. Bar = 2 cm.

1A, B and C), each containing relatively dense stands of *Santalum acuminatum* (R. Br.) A. DC. (Santalaceae) along with a range of woody shrubs and trees. The three sites had not been burnt for at least 5 years prior to the seasons of investigation.

Examination of *S. acuminatum* across the study region during 1993–1996 showed it to spread clonally by prolific

root suckering. Resprouting after fire occurred from bases of burnt shoots (Fig. 1D) and from new suckers forming at intervals along the lateral root stocks of a parent clone (Fig. 1E). Evidence was obtained of systemic death of whole clones after particularly intense fires. Recruitment from seed was very rarely observed and mortality rates of seedlings in nature were suspected to be high.

Species acting as principal hosts for the parasite were identified during the growing seasons (April–October) of 1995 and 1996 by means of a series of shallow excavations around potential hosts and examination of the partly exposed roots for haustoria. Common species whose roots were regularly or occasionally parasitized by *Santalum* at the first site (A) were *Acacia rostellifera* Benth. (Mimosaceae), *Acacia pulchella* R. Br. (Mimosaceae), *Baeckea tetragona* Benth. (Myrtaceae), *Cryptandra mutila* Risseck. (Rhamnaceae), *Melaleuca viminea* Lindley (Myrtaceae), *Alyogyne hakeaefolia* (Giord.) Alef. (Malvaceae), *Anthocercis viscosa* R. Br. (Solanaceae), *Spyridium cordatum* (Turcz.) Benth. (Rhamnaceae), *Exocarpos sparteus* R. Br. (Santalaceae) and *Dodonaea aptera* Miq. (Sapindaceae). At the second site (B) hosts were identified as *Allocasuarina campestris* (Diels) L. Johnson (Casuarinaceae), *Acacia pulchella* R. Br. (Mimosaceae), *Melaleuca viminea* Lindley (Myrtaceae), *Spyridium cordatum* (Turcz.) Benth. (Rhamnaceae), *Leucopogon ovalifolius* Sonder. (Epacridaceae), *Lysinema ciliatum* R. Br. (Epacridaceae), *Desmocladius asperus* Nees (Restionaceae), *Hemiandra pungens* R. Br. (Lamiaceae), *Conostylis candicans* Endl. (Haemodoraceae) and *Gompholobium tomentosum* Labill. (Fabaceae). At the third site (C) parasitized species were *Acacia pulchella* R. Br. (Mimosaceae), *Acacia rostellifera* Benth. (Mimosaceae), *Melaleuca viminea* Lindley (Myrtaceae), *Phyllanthus calycinus* Benth. (Euphorbiaceae), *Allocasuarina campestris* (Diels) L. Johnson (Casuarinaceae), *Desmocladius asperus* Nees. (Restionaceae) and *Hemiandra pungens* R. Br. (Lamiaceae).

Distribution of above ground biomass amongst the above named species was assessed in September 1996 using five randomly selected line transects (100 m long, 1 m wide). Numbers of plants of each species falling within the confines of each 100 m² plot transect were recorded, and mean dry weights of shoot biomass of each species determined from samples of 20 randomly harvested plants of each species from each site. Above ground biomass (dry matter kg ha⁻¹) for each major host was then obtained by multiplying the average shoot dry weight of a species by the mean frequency of occurrence of that species within the transects at a site. Excavations were also undertaken on the root systems of the 20 host species whose above ground biomass had been sampled and each plant scored according to whether or not its roots carried effectively-penetrating haustoria of *Santalum* (see Figs 1 F, 4 A–C).

A simple rating expressed as an ‘accessibility index’ of each species as host for *Santalum* was then computed as the proportion of the above ground biomass of the habitat occupied by the species multiplied by the percentage of plants of that species bearing evidence of effective parasitism by *Santalum*.

An alternative and simpler approach, aimed to provide semi-quantitative ratings for parasite and host densities on a larger scale, utilized colour photographs taken from high elevation positions close to the three sites studied (e.g. see Fig. 1 A, B and C). The enlarged colour prints were then scanned in terms of percentage of projected ground cover occupied by major host species (*Acacia rostellifera*, *A. pulchella* and *Allocasuarina campestris*) using a power

Macintosh 7100/66 computer and the public domain NIH image analyses programme developed at the U.S. National Institute of Health (see Tennakoon and Pate, 1997).

Anatomical studies

The gross morphology of the haustoria and anatomical organization of its interface with host xylem were studied using mature haustoria of *Santalum acuminatum* established on lateral roots of the principal host *Acacia rostellifera*. Low and high resolution light microscope pictures were then taken of longitudinal and transverse sections through the junction region between host and haustorium. Small tissue portions of the interface region of host xylem and endophyte of the haustorium were then excised and fixed in 2.5% glutaraldehyde in 0.025 M phosphate buffer (pH 7.0) and dehydrated and embedded in glycol methacrylate as described by O’Brien and McCully (1981). Transverse sections (2–3 µm thick) through this interface region were prepared, stained with 0.05% toluidine blue (pH 4.2) and examined by high resolution light microscopy (see Pate *et al.*, 1990b).

Sampling of plant material for assays of $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ of plant dry matter

Twenty replicate samples of biomass representative of new season leafy shoot material of each species were collected in spring from within the designated (100 m²) plot transects at each site (site A sampled in August 1995 and sites B and C in September 1996). The material was oven dried at 80 °C for 72 h, ground and then sieved through a 1.25 mm-diameter mesh sieve. Samples (2–3 mg d. wt) of leaves were analysed for $\delta^{15}\text{N}$ and $\delta^{13}\text{C}$ using a SIRA 9 mass spectrometer (VG Isogas, Cheshire, UK) as described by Pate, Stewart and Unkovich (1993) and Tennakoon and Pate (1996a).

Gas exchange measurements

Five individual plants of *Santalum acuminatum* at site A and a matching set of five near-neighbouring plants of the major host *A. rostellifera* were involved in this study. Assimilation rates (net photosynthesis µmol CO₂ m⁻² s⁻¹) and transpiration rates (mmol H₂O m⁻² s⁻¹) of mature foliage of the five pairs of host and parasite were measured on different occasions in spring (September 1995), summer (January 1996) autumn (April 1996) or winter (July 1996), using a portable open cuvette gas exchange system (ADC, model LCA-2, Hoddenson, UK). Measurements were restricted to non-limiting light conditions (1000–1500h) over the 2 consecutive days of each visit to the study site. Each gas exchange measurement took place during a 40–90 s enclosure of the canopy leaves of a species, taking care to avoid periods of rapidly changing light conditions before or during measurements. At the end of the study leaf material of host and parasite used for the measurements was

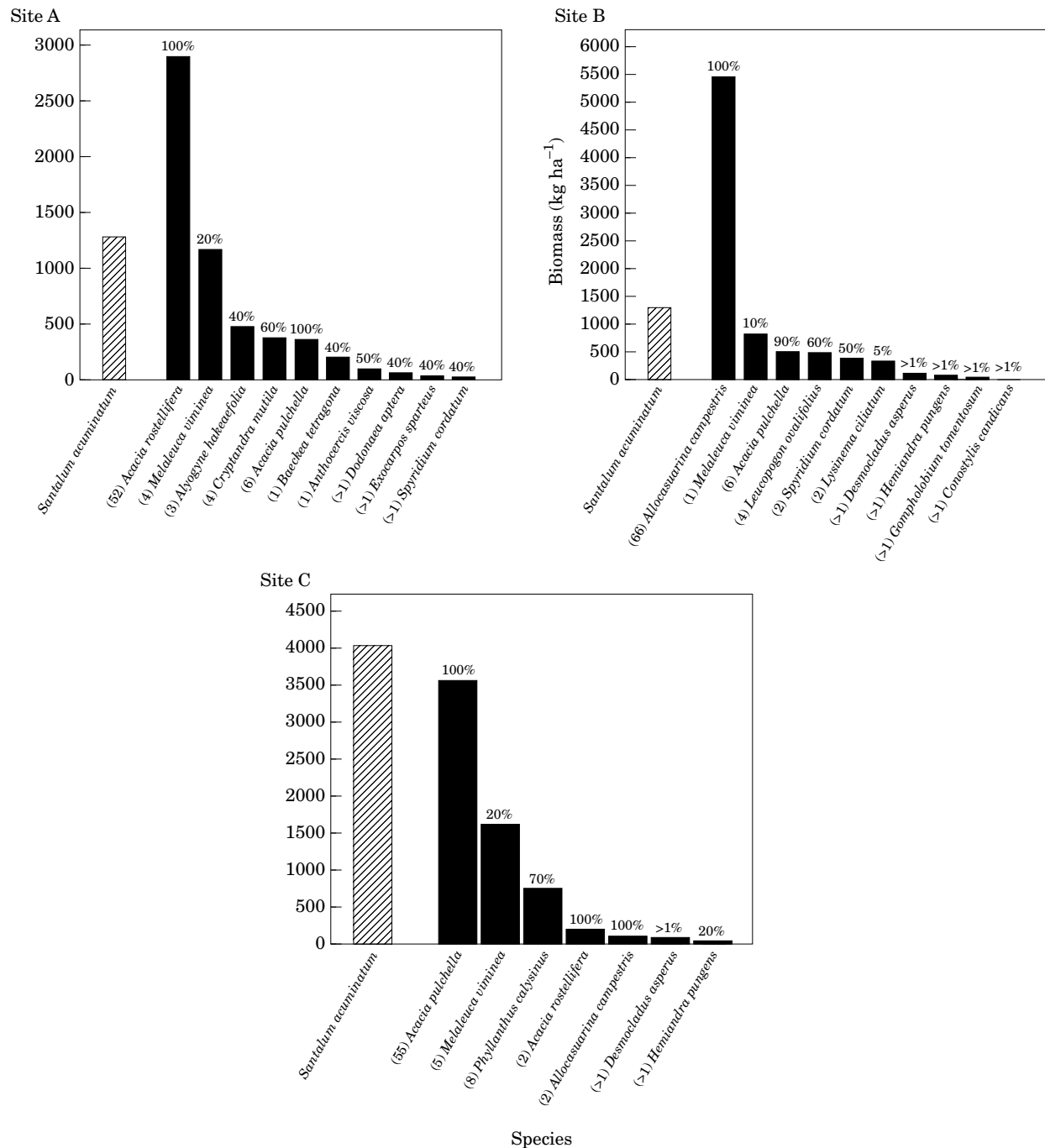


FIG. 2. Biomass distribution of *Santalum acuminatum* and a range of host species from the three sandplain sites investigated. Percentage parasitism recorded for root systems of 20 individuals of each host are given on top of the column bars. Host accessibility indices (as defined in text) are given within brackets against each entry for a plant species.

harvested, dried and subjected to ¹³C analyses as described above.

Measurement of specific hydraulic conductivity (K_s) of roots of *Santalum* and *Acacia rostellifera*

This study used major lateral roots of *Santalum* and of its predominant host *A. rostellifera* excavated randomly from five different locations at the study site A. Root segments

20 cm long were excised from 25 *Santalum* plants and from parasitized (1 or more *Santalum* haustoria) and unparasitized lateral roots of a set of 25 matching plants of *A. rostellifera*. The cut ends of all segments were shaved smooth with a fresh razor blade, and the material immediately taken back to the laboratory whilst still fully submerged in water. Stele segments were then prepared for measurements of specific hydraulic conductivity (K_s) essentially as described by Tennakoon and Pate (1996b). The

methodologies used were as described by Gibson, Calkin and Nobel (1984) and Schultz and Matthews (1993), and as adapted especially for roots by Pate, Jeschke and Aylward (1995).

Root segments used for conductivity measurements were vacuum perfused for 1 h with a filtered solution of 0.1% basic fuchsin in order to distinguish presumptive conducting xylem elements from fibres and other non-conducting tissues. Transverse sections were then cut mid-way through each segment using a wood microtome and enlarged colour photomicrographic images of these used to measure the proportion of the total transectional areas of segments comprised of the lumina of stained xylem vessels (see Pate *et al.*, 1995). It was then possible to convert values for specific hydraulic conductivities based on mean stele transectional area to those computed in terms of transectional areas of conducting xylem tissue.

RESULTS

Biomass distribution of hosts and parasites and extents of parasitism of different hosts by Santalum

Based on their respective proportional distributions in total above ground biomass (Fig. 2), percentage ground cover (Fig. 3), or respective rankings in terms of the host accessibility index (% values given within brackets against

the entry of host species in Fig. 2), *Acacia rostellifera* turned out to be the predominant host to *Santalum* at site A, *Allocasuarina campestris* the principal host at site B, and *Acacia pulchella* the most heavily parasitized species at site C. High levels of parasitism were observed on virtually all plants of these alternative major hosts and all mature haustoria sampled from the species showed successful penetration to the host xylem.

Herbaceous species at the three study sites were mostly unparasitized presumably due to the fine and ephemeral nature of their roots. Amongst the woody species, a particularly abundant shrub *Melaleuca viminea* was poorly parasitized, presumably due to ineffective establishment of haustoria on the continuously exfoliating periderm of roots of this myrtaceous species. Medium levels of parasitism by *Santalum* were observed on lateral roots of another root hemiparasite *Exocarpos sparteus* at site A, but no examples of reverse parasitism of *Santalum* by *Exocarpos* were evident. The presence of mantles of sand grains trapped within the persistent root hairs of Cyperaceae and a range of other monocotyledons (Pate and Dixon, 1996) encountered in the habitats was regarded as playing a vital protective role against invasion by *Santalum*.

Santalum was the largest single component in terms of biomass (41%) at site C where its principal host was *Acacia pulchella* (Fig. 2C). At site A, 23% of the above ground biomass was comprised of *Santalum*, compared with 17% at site B. Corresponding larger scale photographically-based estimates of percentage ground cover occupied by *Santalum* and its principal hosts (Fig. 3) agreed reasonably well with respective rankings based on line transects at the three sites, but suggested greater proportional ground cover of *Santalum* at sites C and B than A.

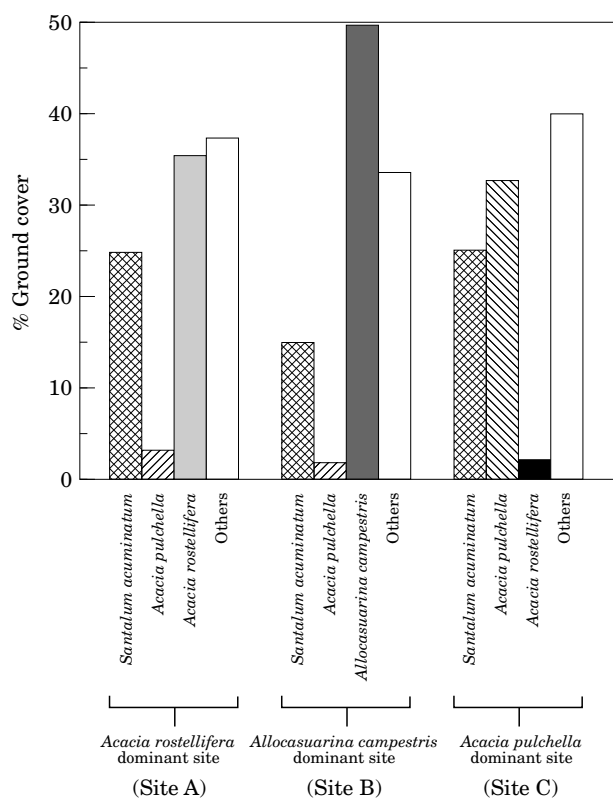


FIG. 3. Percentage ground cover occupied by *Santalum acuminatum* and previously determined three major host species in each of the three sites investigated.

Ontogeny of haustoria

Regardless of the host parasitized, haustoria of *Santalum acuminatum* exhibited the bell-shaped configuration typical of most other Santalalean taxa. Each haustorium tapers proximally by a narrow stalk joining on to its parent root (Figs 1F, 4A). Regular monitoring of the relative proportions of young, mature and senescent haustoria of *Santalum* on commonly parasitized hosts (*Acacia rostellifera* and *A. pulchella*) over the period October 1994 to October 1996 (data not shown) indicated that haustorial establishment and penetration took place mostly in autumn, that appreciable proportions of the haustorial population senesced during high summer but some survived into the next growing season or at least until the new season's crop of haustoria had established in the following autumn. Major roots of hosts often showed scars where haustoria were attached in previous years (cf. similar observations on *Olex* recorded by Pate *et al.*, 1990c).

The gross morphology of haustoria (Figs 1F, 4A, B and C) showed the appressorium and associated cortical fold of the haustorium to only partly encircle the host root. Penetration to the cambium was accomplished by means of a wedge-shaped 'sucker' (endophyte) whose long axes lay parallel with that of the host root. On reaching the cambium,

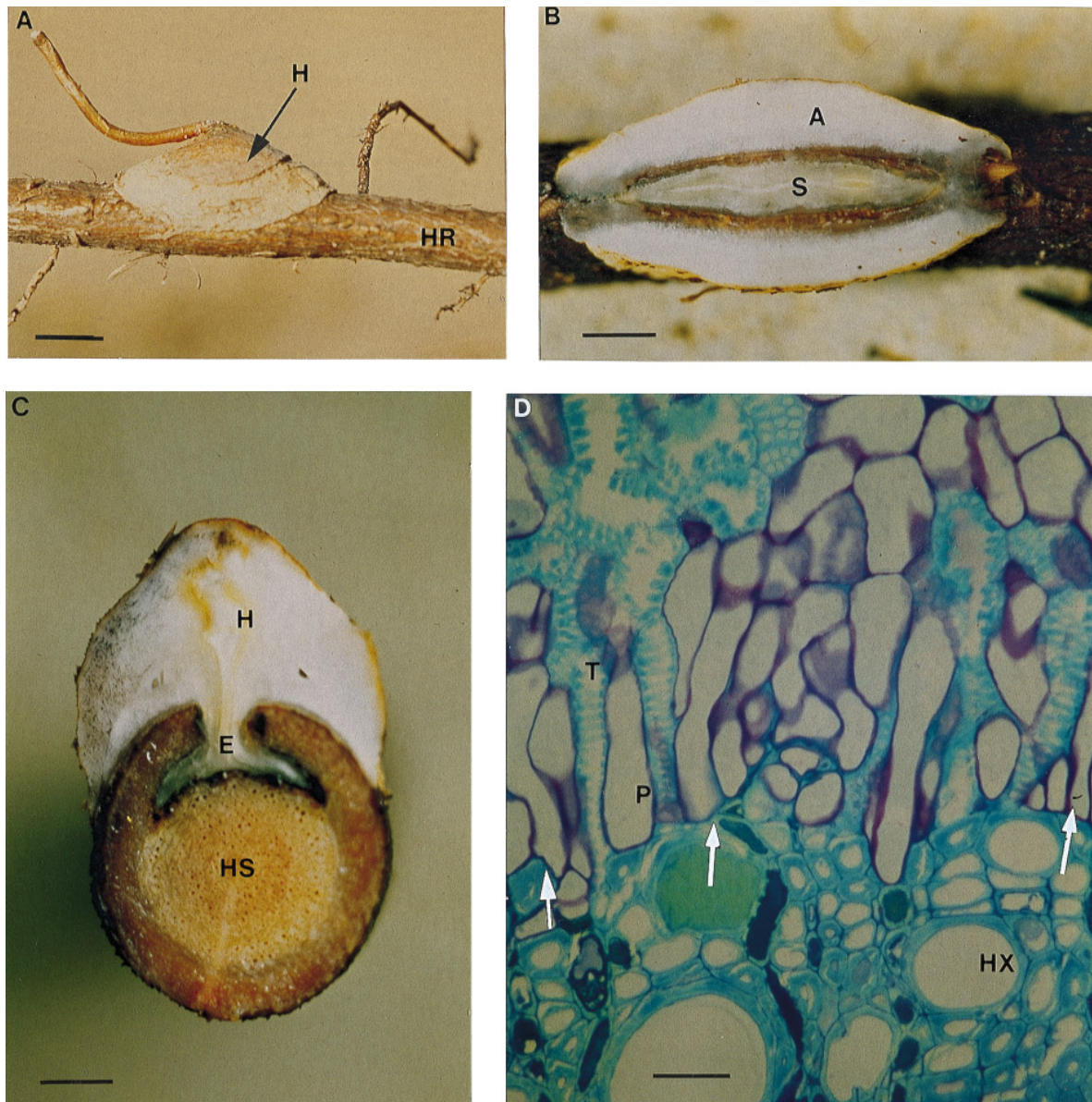


FIG. 4. Photographs of whole haustoria or transverse sections of haustoria of *Santalum* when attached to its common host *Acacia rostellifera*. A, Single haustorium (H) on a lateral root of *Acacia rostellifera* (HR). Bar = 1 cm. B, Median section taken parallel to the host-parasite interface. The original position of the appressorial disc of the haustorium (A) and the haustorial sucker (endophyte) penetrating to the host xylem (S) can be seen. Bar = 0.75 cm. C, Transverse section of mid region of a haustorium (H) of *Santalum acuminatum*. The endophytic tissue (E) of the haustorium has penetrated the cortex of the host root (*A. rostellifera*) to establish direct contact against the host stele (HS). Bar = 0.4 cm. D, Detailed anatomy of the haustorial interface between *S. acuminatum* and *A. rostellifera*. Demarcation between parasite and host root xylem tissue (HX) is marked by arrows. Note the thick walled contact parenchyma (P) and terminating file of tracheary elements (T) in the endophyte and absence of direct lumen-to-lumen tracheary contact between host and parasite. Bar = 100 μ m.

the sucker flattened out laterally to form an ellipsoidal disc of endophytic tissue closely pressed against the exposed host xylem (Fig. 4B and C).

Anatomy of the host-parasite interface (Fig. 4D) showed a marked preponderance of haustorial parenchyma as opposed to terminating tracheary elements adjoining the host xylem. Direct lumen-to-lumen tracheary continuities between xylem of host and parasite were not encountered in any of material examined.

$\delta^{15}\text{N}$ values of species encountered in the study sites

Mean $\delta^{15}\text{N}$ values for newly formed leafy shoot material collected from *Santalum* and each identified host species in the spring of 1995 or 1996 were as shown in Fig. 5. Shoot material from sites A and C showed mostly positive $\delta^{15}\text{N}$ values, whereas that from site B consistently recorded negative values (Fig. 5B). The reason for this difference between sites was not apparent. As reported earlier for

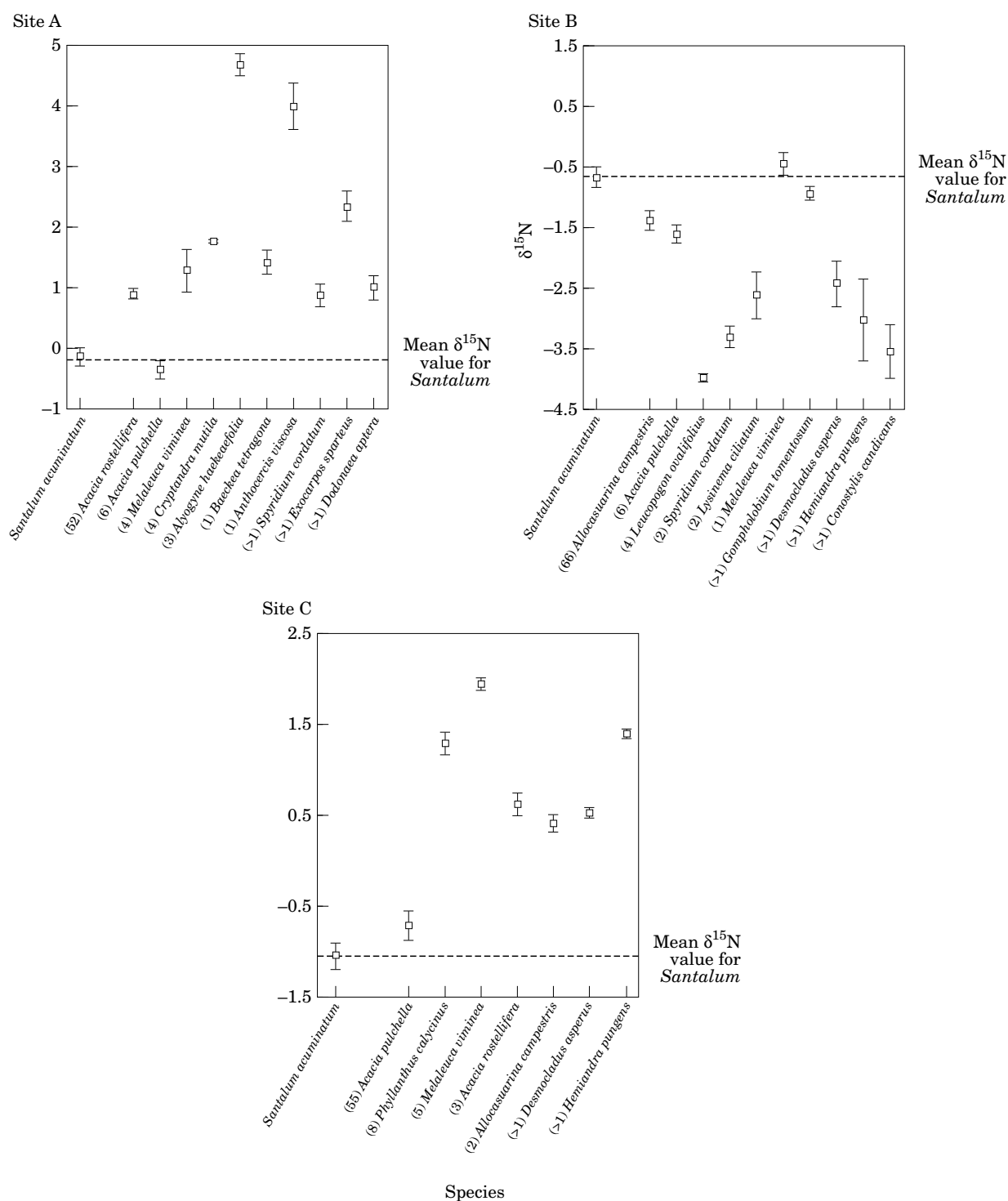


FIG. 5. Mean $\delta^{15}\text{N}$ values for dry matter of new season leafy shoot material of *Santalum* and principal hosts at the three sites investigated. Values are given as means ($\pm$ s.e.) of analyses of dry matter samples from 20 replicate plants of each species. Host availability indices (as defined in the text) are given against each entry.

other West Australian habitats (Pate *et al.*, 1993), nitrophillous species such as *Alyogyne hakeaefolia* and *Anthocercis viscosa* and other mostly mycorrhizal non- N_2 -fixing species all showed $\delta^{15}\text{N}$ values diverging markedly from those of putative N_2 -fixing species. The latter (*Acacia*

rostellifera, *A. pulchella*, *Gompholobium tomentosum* and *Allocasuarina campestris*), all major hosts to *Santalum* at the sites, showed mean $\delta^{15}\text{N}$ values much closer to that of atmospheric N_2 (0‰) than most other non-fixing hosts (Fig. 5), suggesting heavy (70% or more) dependence on

TABLE 1. Gas exchange characteristics and $\delta^{13}\text{C}$ values of canopy leaves of the root hemiparasite *Santalum acuminatum* and commonly parasitized host *Acacia rostellifera*

	September 1995		January 1996		April 1996		July 1996	
	<i>Santalum</i>	<i>Acacia</i>	<i>Santalum</i>	<i>Acacia</i>	<i>Santalum</i>	<i>Acacia</i>	<i>Santalum</i>	<i>Acacia</i>
Photosynthesis* ($\mu\text{mol m}^{-2} \text{s}^{-1}$)	3.05 $\pm$ 0.27	7.71 $\pm$ 0.32	3.14 $\pm$ 1.50	9.54 $\pm$ 0.53	1.48 $\pm$ 0.34	5.03 $\pm$ 0.45	2.88 $\pm$ 0.17	4.25 $\pm$ 0.30
Transpiration* ($\text{mmol m}^{-2} \text{s}^{-1}$)	0.91 $\pm$ 0.10	1.57 $\pm$ 0.07	1.50 $\pm$ 0.13	3.65 $\pm$ 0.22	1.08 $\pm$ 0.23	2.13 $\pm$ 0.22	1.06 $\pm$ 0.10	1.52 $\pm$ 0.11
Water use efficiency ($\text{mmol CO}_2 \text{ mol H}_2\text{O}$)	3.72 $\pm$ 0.62	4.98 $\pm$ 0.23	2.21 $\pm$ 0.14	2.71 $\pm$ 0.13	1.34 $\pm$ 0.07	2.40 $\pm$ 0.14	2.88 $\pm$ 0.24	2.90 $\pm$ 0.27
$\delta^{13}\text{C}$	-28.6 $\pm$ 0.1	-28.7 $\pm$ 0.2	-27.6 $\pm$ 0.5	-28.1 $\pm$ 0.3	-27.6 $\pm$ 0.3	-28.1 $\pm$ 0.1	-27.3 $\pm$ 0.1	-28.3 $\pm$ 0.6

Values as means $\pm$ s.e. of five different plants in the natural habitat.

* Measurements were taken over 2 d (1000–1500 h) at an ambient CO_2 concentration of approx. $350 \mu\text{mol mol}^{-1}$, relative humidity 40–45% and when photosynthetic photon flux density (PPDF) was between 1000 and $1300 \mu\text{mol m}^{-2} \text{s}^{-1}$ in *Santalum* and targeted *Acacia* plants within the root catchment of the root hemiparasite.

symbiotically fixed N (see Shearer and Kohl, 1986; Handley and Raven, 1992). Consistent with this, the latter hosts were consistently well nodulated at each site in which they occurred.

Santalum recorded a mean $\delta^{15}\text{N}$ value of $-0.13 \pm 0.01\text{‰}$ at site A where $\delta^{15}\text{N}$ values of the major hosts *Acacia rostellifera* and *A. pulchella* were 0.89 ± 0.08 and $-0.35 \pm 0.10\text{‰}$, respectively. The corresponding value for *Santalum* at site B was $-0.64 \pm 0.15\text{‰}$, compared with a $\delta^{15}\text{N}$ value of $-1.34 \pm 0.15\text{‰}$ for the major host *Allocasuarina campestris*. At site C the *Santalum* value was -1.04 ± 0.15 against a $\delta^{15}\text{N}$ value of $-0.69 \pm 0.16\text{‰}$ for the major host, *Acacia pulchella*. The similarity between these $\delta^{15}\text{N}$ values for *Santalum* and those of the major N_2 fixing hosts at the three sites suggest that the parasite mainly relies on fixed N as opposed to direct uptake of soil inorganic N or acquisition of soil derived N from non-fixing hosts other than legumes and *Allocasuarina*.

Gas exchange characteristics and $^{13}\text{C}/^{12}\text{C}$ isotope ratios of *Santalum acuminatum* and its common hosts

A comparative study of day time gas exchange of leaves of *Santalum* and near-neighbouring plants of *Acacia rostellifera* at site A was conducted in spring (September 1995), mid summer (January 1996), autumn (April 1996) and winter (July 1996). The data (Table 1) collectively indicated that *Santalum* consistently transpired and photosynthesized at much lower rates than its host. Water-use efficiencies were similar between the parasite and host on each study occasion. $\delta^{13}\text{C}$ values of the actual canopy leaves involved in the gas exchange measurements (Table 1) showed close similarity for host and parasite, and there was no evidence of obvious drift in $\delta^{13}\text{C}$ signals with season. The data are consistent with the water-use efficiency of *Santalum* being similar to that of the host *Acacia rostellifera* (see Farquhar and Richards, 1984).

A broadly based comparison of $\delta^{13}\text{C}$ signals of parasites and hosts was also conducted on the same bulk shoot material utilized for the above mentioned $\delta^{15}\text{N}$ measure-

ments. The results (Fig. 6) showed *S. acuminatum* from site A to have above ground dry matter with more negative $\delta^{13}\text{C}$ values (-27.6‰) than its principal hosts at site A. At site B the value for *Santalum* (-26.8‰) was less negative than that of most hosts, whereas at site C *Santalum* (-27.6‰) was more negative than all but one of its hosts. With such unexplainable inter-site variability and lack of information on seasonal trends in $\delta^{13}\text{C}$ signals of species, it would be unwise to use the data to assess water use efficiencies and extents of water stress of *Santalum* relative to its hosts.

Specific hydraulic conductivities and mean vessel radii of lateral roots of *Santalum* and a major host *Acacia rostellifera*

Data for specific conductivities (K_s) and anatomical features of lateral roots of *Santalum* and of parasitized and unparasitized *Acacia rostellifera* are summarized in Fig. 7. Specific conductivity (stele area basis) values for both parasitized and unparasitized *Acacia* roots across the five sampling locations were consistently several fold greater than those of *Santalum* (Fig. 7A). Mean radii (Fig. 7B) of conducting elements of parasitized and unparasitized *Acacia* lateral roots were also three or more times higher than for comparable *Santalum* roots, whereas proportions of root stele occupied by lumina of conducting elements of xylem (Fig. 7C) were very similar between parasite and the two types of host root. As expected of its wider vessels, K_s values computed on a xylem lumina area basis proved to be higher in both classes of root of *Acacia* than of *Santalum* (Fig. 7D). However, the three- to eight-fold higher K_s (xylem lumina area basis) of *Acacia* compared to *Santalum* was less than anticipated from the r^4 -based increase in K_s predicted from the Hagen-Poiseuille equation on flow performance within smooth walled capillaries (see also Pate *et al.*, 1995). Anatomical investigations showed tyloses to be absent from conducting sap wood of both *Acacia* and *Santalum* and general anatomical comparisons of parasitized and unparasitized *Acacia* roots indicated no obvious effects of the presence of haustoria on the structure of host roots.

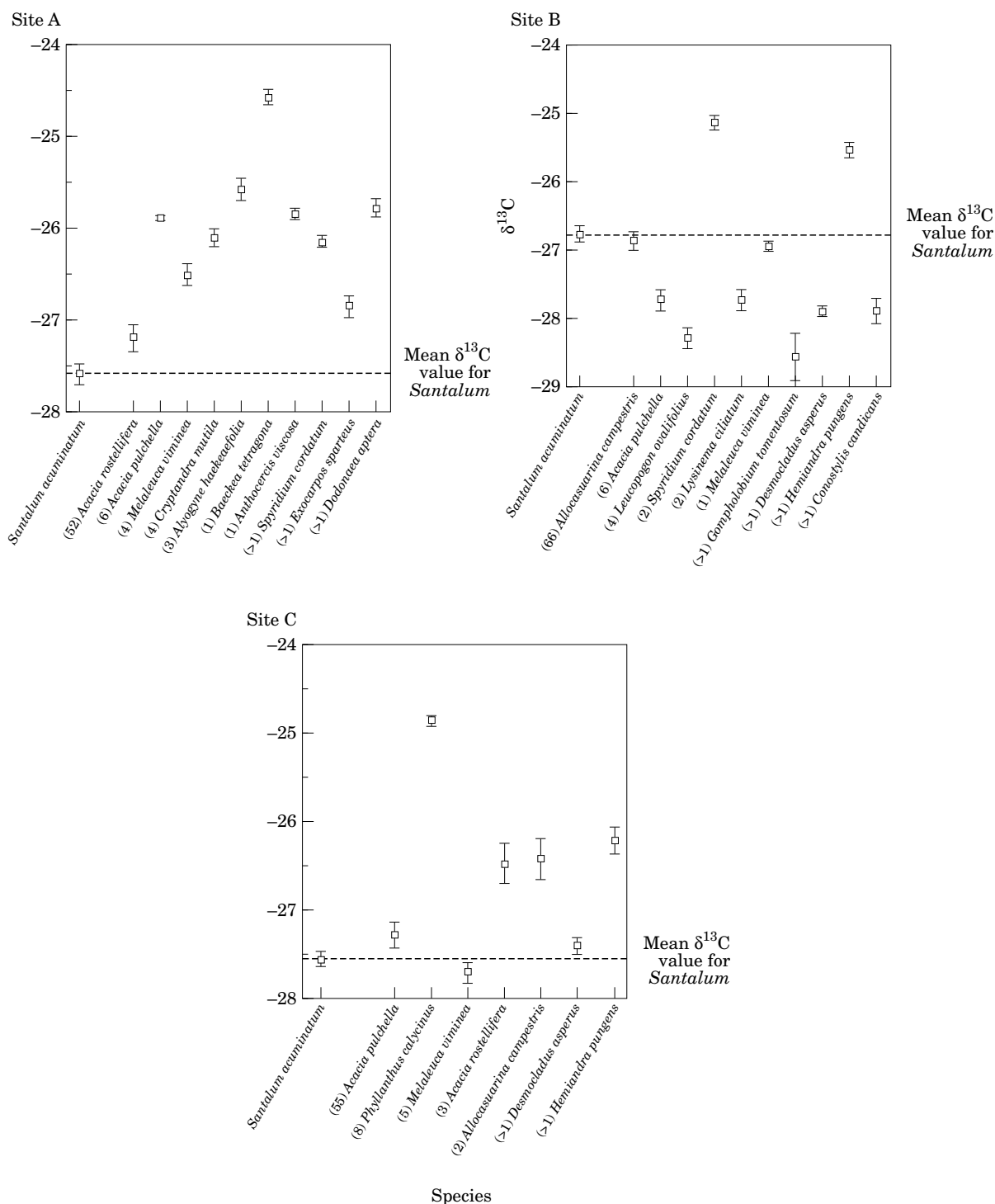


FIG. 6. Mean $\delta^{13}\text{C}$ values for dry matter of new season leafy shoot material of *Santalum* and its major hosts at the three sites investigated. Values are given as means ($\pm$ s.e.) of analyses of dry matter samples from 20 replicate plants of each species.

DISCUSSION

This paper examines a number of interrelated biological and physiological features which might be considered relevant to the complex interaction of the quandong (*Santalum acuminatum*) and its commonly exploited hosts in coastal heathlands of south-western Australia. Above ground

biomass densities, percentage ground cover values and haustorial frequencies on hosts at each of three contrasting sites indicated a strong preference of the parasite for potential N_2 fixing species. This finding is in agreement with the already-reported host ranges for other root hemiparasites such as *Olax phyllanthi* (Pate *et al.*, 1990c; Tennakoon and Pate, 1996a), *Colpoos compresum* (Fineran,

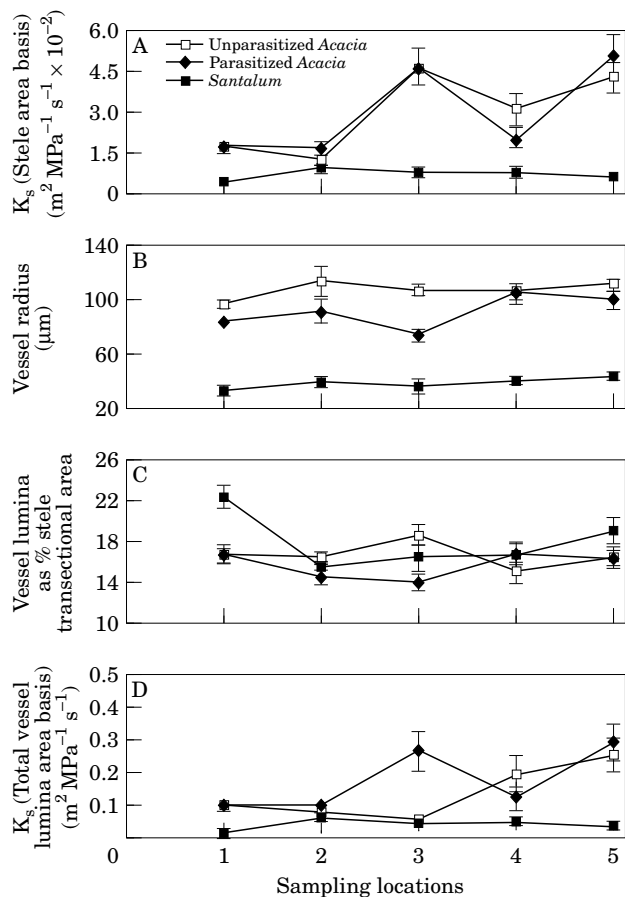


FIG. 7. Relationships between conductivity based on stele transectional area (A), mean vessel radius (B), proportion of stele transectional area occupied by vessel lumina (C) and conductivity based on total vessel lumina area basis (D) in *Santalum acuminatum* and parasitized and unparasitized *Acacia rostellifera*. Data are given for each of five sampling locations (1–5) at study site A.

1994), *Rhinanthus minor* and *Euphrasia frigida* (Gibson and Watkinson, 1989; Seel and Press, 1993; Press and Seel, 1996), which have been shown to selectively parasitize leguminous species in their respective habitats. In such instances one would predict that a parasite, directly abstracting xylem-derived nitrogenous compounds from N_2 fixing hosts, would be at a selective advantage over companion non-fixing, non parasitic species entirely dependent on soil nitrogen. Such advantages might be especially important under the markedly N deficient S. W. Australian heathland soils typical of the present study on *Santalum* and the companion one on *Olax*, and indeed might explain why woody root hemiparasites may become the dominant growth forms and occupy unusually large proportions of the resident biomass of such ecosystems.

The close to zero $\delta^{15}\text{N}$ values for shoots of the *Acacia*, *Allocasuarina* and *Gompholobium* hosts as opposed to values diverging markedly from zero for other non-fixing host species suggest that the former species were obtaining a significant fraction of their N from the atmosphere at all of the study sites investigated (Handley and Scrimgeour, 1997). In addition, shoot $\delta^{15}\text{N}$ values of *Santalum* from each study site were closest to those of major N_2 fixing hosts suggesting

the *Santalum* is in turn drawing upwards of 70% of its N from these hosts.

Just as recorded earlier in our parallel studies on *Olax phyllanthi* (see Pate *et al.*, 1990c), maximum numbers of mature haustoria exist on hosts in spring and early summer when growth and nutrient acquisition by species are likely to be maximal. Survival of small numbers of haustoria through the summer and at least until the next generation of haustoria forms the following autumn, would then ensure effective continuity with hosts in terms of abstraction of water and nutrients.

Low levels of parasitism by *Santalum* on almost all herbaceous species encountered at the study sites can be partly attributed to the ephemeral nature and relatively narrow diameters of the roots of such hosts compared with those of the more favoured woody species. However, there may still be specific biochemically-driven mechanisms positively selecting woody species, especially in relation to the consistently intensive exploitation of N_2 fixing shrubs and trees such as *Acacia rostellifera* and *Allocasuarina campestris*. Nevertheless, judging from the findings that *Santalum* has the ability to parasitize a very broad range of host taxa, specific host-derived signals similar to those reported to initiate haustorial formation in other host specific root hemiparasites (see Steffens *et al.*, 1982; Chang and Lynn, 1986; Netzly *et al.*, 1988; Ayres, Press and Spencer-Phillips, 1996) appear to be unlikely in *Santalum*. In any event it would obviously be highly advantageous to shallow, laterally extensively rooted species such as *Santalum* and *Olax* to attach to many woody host species, especially including some deep rooting hosts. As shown elsewhere for *Olax* (Pate *et al.*, 1990c), parasitism of the latter would allow the parasite to tap water during the severe summer droughts typically experienced in the sandplains of south western Australia.

Anatomically the haustorium of *Santalum* resembles that of *Olax* (Pate *et al.*, 1990c; Tennakoon and Pate, 1997) and *Osyris* (Niranjana and Shivamurthy, 1987) in possessing an endophyte which spreads laterally around the surface of the host xylem. This contrasts with the deeper penetration or complete splitting of host xylem observed for endophytes of *Thesium* (Weber, 1977), *Atkinsonia* (Fineran and Ingerfeld, 1982) and *Nuytsia* (Fineran, 1983). In all cases examined here for *Santalum*, direct luminal continuities between the conducting elements of the xylem of the host and parasite are absent, just as recorded for *Olax* (Pate *et al.*, 1990b; Tennakoon *et al.*, 1997a), implying that mass tracheary flow of water and nutrients from hosts to *Santalum* is likely to be non-existent or severely restricted. This contrasts with the situation in mistletoes (Pate, True and Kuo, 1991) in which lumen-to-lumen pathways have been observed between host and parasite. One would then view the largely parenchymatous surface of the haustorium of parasites such as *Santalum* and *Olax* as being principally involved in selective uptake and processing of host xylem solutes as suggested in our companion study on *S. acuminatum* (Tennakoon *et al.*, 1997) and elsewhere in our studies of *Olax phyllanthi* (Pate *et al.*, 1994; Tennakoon and Pate, 1996a, 1997).

The relatively low rates of day time transpiration and photosynthesis shown by *Santalum* in comparison with its

hosts is another point of similarity to the behaviour of *Olex* (see Pate *et al.*, 1990a). However, the relatively low transpiration rates and good water use efficiencies shown for *Santalum* and *Olex* contrast markedly to published data on mistletoes and herbaceous root hemiparasites (see Schulze, Turner and Glatzel, 1984; Press, Tuohy and Stewart, 1987; Press, Graves and Stewart, 1988; Davidson, True and Pate, 1989; Press, Smith and Stewart, 1991; Küppers *et al.*, 1992). In all of these latter cases, transpiration of the parasitic partner was much higher and water-use efficiencies considerably poorer than those of hosts.

Although the gas exchange data of this study were limited to occasional day time measurements during the four seasons, and involved *Santalum* in association with only one host, the data obtained correlate well with carbon isotope discrimination values for dry matter of the canopy leaves used for measurements. Thus, assuming as suggested by Farquhar and Richards (1984) that differences in $\delta^{13}\text{C}$ values between host and parasite would directly reflect differences in water use efficiencies, the data would imply essentially no differences in water use efficiencies between *S. acuminatum* and *A. rostellifera*. This is consistent with the above mentioned gas exchange data. However, large inter-site differences in $\delta^{13}\text{C}$ values of the bulk shoot dry matter of *Santalum* and its principal hosts in the study region would indicate a considerably more complex situation than suggested above for *Acacia* and *Santalum*. The reasons for these substantial inter-site differences are not clear, but previously demonstrated variation in $\delta^{13}\text{C}$ values of plant dry matter related to organ type, age, composition and even the position on the plant (e.g. Winkler *et al.*, 1978; Benner *et al.*, 1987; Boutton, 1990) may well account for much of the variability obtained. Use of $\delta^{13}\text{C}$ values for assessing the extent of seasonal water stress or water-use efficiencies of hosts and parasites would clearly be unwarranted in the circumstances.

The low rates of photosynthesis shown for *Santalum* in the gas exchange study suggest that the parasite may be depending heavily on its hosts for carbon. This conclusion is in agreement with our accompanying paper (Tennakoon, Pate and Stewart, 1997) in which it was concluded that approximately one third of the C of the dry matter gain of the parasite was likely to be derived from heterotrophic intake of organic solutes from hosts. In view of the low transpiration rates and lack of luminal continuities between conducting elements of the host and *Santalum*, bulk mass intake of xylem fluids from a host would be expected to play a minimal role in solute acquisition. However, we have suggested (Tennakoon *et al.*, 1997b) that proline synthesized in the haustoria and exported to the shoot of *Santalum* might contribute significantly in the generation and maintenance of osmotic gradients between host and parasite, and thus facilitate efficient host-derived solute transfer to *Santalum* in the absence of substantial influx of xylem water. Results of this nature put a new perspective on the role of the haustorium in parasite functioning.

As in the case of other root hemiparasites, water movement through the soil-plant-atmospheric-continua of *Santalum* and its hosts is likely to be highly complex, due to the possibilities of hosts supplying water in differing relative

amounts or even withdrawing water from the parasite via haustoria and *Santalum* meeting substantial proportions of its water requirement by direct uptake from soil via its own root system. Our present comparison of specific hydraulic conductivities of roots of *Santalum* and *Acacia rostellifera* provide somewhat limited information on this issue except to demonstrate substantially greater resistance to water flow through roots of host and parasite. While this phenomenon might be viewed as further evidence of conservatism in water use by the parasite, just as suggested from the gas exchange data, the significance of the finding cannot be properly evaluated until more comparisons with other hosts are undertaken and data considered in relation to rooting patterns and relative demands of the parasite and host for water. Furthermore, it has not yet been possible to examine the resistance offered by the haustorium so we have little information on its overall impact on intake of water from hosts. Aspects of this nature have yet to be properly considered for any root hemiparasite.

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